

WELL COMPLETION REPORT (SYNTHETIC)

OIL-NHK-004 - Nahorkatiya-004

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000038
Field / District	Nahorkatiya / Dibrugarh
Block	PEL-ASSAM-02
Basin	Assam-Arakan
Formation	Tipam (TIP)
Spud / Completion	1999-05-30 / 1999-08-14
Measured / TVD	2857 m / 2838 m
Trajectory	Deviated; inc 14.8 deg; az 56.8 deg
Formation Interval	1522-2578 m
Lithology	Fluvial sandstone with clay streaks
Pore Pressure Reference	3210 psi
Fracture Gradient	14.33 ppg
Mud System	PHPA Water-Based Mud
NWIS Well Risk	Moderate (47.9/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	714	Surface
9 5/8 in casing	1657	Intermediate
7 in liner/casing	2514	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
1999-06-18 866 m	Mud Loss	Critical	Intermittent circulation losses developed while drilling ahead. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Track flow balance and pit volume trend before losses become severe.
1999-07-17 1887 m	Torque Spike	Critical	Torque rose without corresponding pressure anomaly. Mitigation: Ream and condition hole if torque remains elevated. Lesson: Avoid simultaneous high WOB and high RPM in correlated interval.
1999-06-18 846 m	Mud Loss	Medium	Intermittent circulation losses developed while drilling ahead. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
1999-06-22 1025 m	Torque Spike	Medium	High torque developed while WOB remained elevated. Mitigation: Reduce WOB/RPM and circulate before resuming normal parameters. Lesson: Avoid simultaneous high WOB and high RPM in correlated interval.
1999-06-24 1064 m	Mud Loss	Medium	Partial returns observed at shale shaker. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
1999-06-28 1221 m	Mud Loss	Medium	Intermittent circulation losses developed while drilling ahead. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
1999-06-28 1240 m	Stuck Pipe	Medium	Rotary movement deteriorated progressively. Mitigation: Condition mud and verify cuttings transport before connection. Lesson: Use offset stuck-pipe depth to increase circulation and reaming frequency.

Completion Summary

The synthetic well reached 2857 m MD in the Tipam interval. The NWIS knowledge record links 12 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.